

Optimized Design of a Stand-Alone Hybrid PV/Wind/Diesel Energy System for Sustainable Development in Sudan: Minimizing Costs and Carbon Emissions

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Abstract— Fossil fuels account for 52% of Sudan's primary energy consumption, while hydropower contributes approximately 42%. As part of its energy strategy, the country aims to increase the share of renewable energy to 11% by 2031. This study focuses on Southern Sudan, which is an ideal case study for renewable energy integration due to its abundant solar radiation and moderate wind speeds. The research investigates the design and optimization of a sustainable, stand-alone hybrid energy system for domestic use, with a focus on minimizing total costs and environmental impact. The study employs real-world data on solar radiation, wind speeds, and other relevant parameters to develop a hybrid optimization model using HOMER and MATLAB software. The system's projected lifetime is 24 years. The primary goal of this paper is to design an optimized hybrid system comprising photovoltaic (PV) panels, wind turbines, diesel generators, and battery storage to minimize the levelized cost of energy (LCOE) and reduce CO₂ emissions. Results indicate that the cost of energy can be reduced by 29%, and greenhouse gas emissions can be decreased by 51% compared to a diesel-only system. By integrating solar and wind energy with diesel backup, the proposed system achieves optimal cost efficiency while ensuring reliable power supply. Furthermore, the system is designed with scalability in mind, enabling future expansion to meet growing energy demand.

Keywords—Hybrid renewable energy, Sustainable energy, optimal cost, fossil fuel, Carbon emission, South region of Sudan HOMER Pro®

I. INTRODUCTION

Electrical energy has become integral to all facets of modern life, leading to an increasing demand for energy production to meet growing consumption needs. In response, efforts have been made to enhance renewable energy capacity, as exemplified by the National Energy System Development (NESD), which encourages the exploration of hybrid energy systems. Due to environmental concerns and the implementation of energy conservation policies, nations around the world are taking steps to reduce greenhouse gas emissions and limit the use of fossil fuels. As a result, renewable energy sources have gained significant attention

over the past few decades, driving the development of integrated systems that combine both renewable and non-renewable resources to address the rising energy demand. This is where the concept of hybrid energy systems emerges [1].

In remote regions, diesel generators are commonly used to supply electrical energy due to their reliability, availability, and ease of operation [2]. However, diesel generators come with significant drawbacks, including high maintenance costs, elevated fuel expenses, and negative environmental impacts, particularly in terms of CO₂ emissions [3]. In contrast, hybrid power systems offer the capability to deliver consistent, high-quality electricity around the clock, while mitigating some of these issues. However, relying on a single renewable energy source, such as solar or wind power, is often insufficient due to high system costs and the need for extensive energy storage [4-8]. To address this, hybrid systems that integrate renewable sources like wind and solar energy with conventional power systems, such as diesel generators and battery storage, offer a viable solution for delivering reliable and sustainable energy.

This hybrid system provides several advantages, including improved efficiency, flexibility in energy planning, and significant environmental benefits such as reducing carbon emissions and minimizing reliance on fossil fuels compared to stand-alone diesel generators [9]. As a result, hybrid systems combining PV, wind, diesel, and battery storage have become increasingly competitive with conventional diesel-only systems [3]. The primary objectives of these hybrid approaches are to mitigate environmental impacts, conserve energy, reduce the depletion of exhaustible resources, and enhance overall system safety.

The utilization of hybrid solar-wind systems is critical for advancing sustainable energy solutions. These systems have been extensively studied to evaluate their development and performance. One approach involves the integration of solar energy with wind turbines, incorporating innovative designs to enhance efficiency and adaptability to varying

environmental conditions. Research on hybrid renewable energy systems (HRES) has also addressed key aspects such as optimal sizing, feasibility analysis, system modeling, control strategies, and reliability, providing valuable insights for improving their performance and deployment [13-17].

Makbul A.M. Ramli [18] et.al, presented case study model on the hybrid solar and wind system on the techno-economic energy analysis for in Saudi Arabia. The study is carried out for economic production for the electric using the hybrid system; the different parameters are taken into consideration for economic production. A hybrid wind and photovoltaic power generation system was designed for a remote area in Ethiopia to address basic electrification requirements. Data for the study was collected from the national agency, and the system was analyzed through simulation using HOMER software. The results demonstrated that the system operates effectively, with the electricity shortage reduced by up to 20%, highlighting its potential for addressing energy challenges in remote regions.

Innovative techniques have been explored for hybrid power generation, incorporating wind energy and photovoltaic (PV) modules to improve efficiency and performance. For instance, one approach utilizes wind energy to cool PV modules, combining Savonius and Darrieus wind turbines with PV systems to enhance overall output. Additionally, optimized hybrid solar and wind power generation methods have been developed, particularly for remote areas, demonstrating significant reductions in coal consumption and carbon emissions during electricity production. Advanced designs of hybrid systems have also employed technologies such as doubly-fed induction generators (DFIG) in wind energy conversion systems (WECS). These systems benefit from the integration of power electronics, which has been shown to significantly enhance power quality and system performance [16-19].

II. SYSTEM CONFIGURATION

This study focuses on the southern region of Sudan (Latitude 15.60° N, Longitude 22.79°, Time zone UT+2, Altitude 354 m), where the existing energy system primarily relies on two diesel generators. These generators contribute significantly to high fuel consumption and have a considerable environmental impact on the surrounding city. The site itself comprises heavy-duty facilities, making it an ideal case for exploring more sustainable energy solutions.

A. Load Profile of the Selected Site:

The region utilizes a diesel engine power plant that operates with crude oil as the primary fuel source. Due to the cold winter conditions, additional power generation is required to heat the crude oil, leading to substantial fuel consumption during the early hours of the day (00:00–06:00).

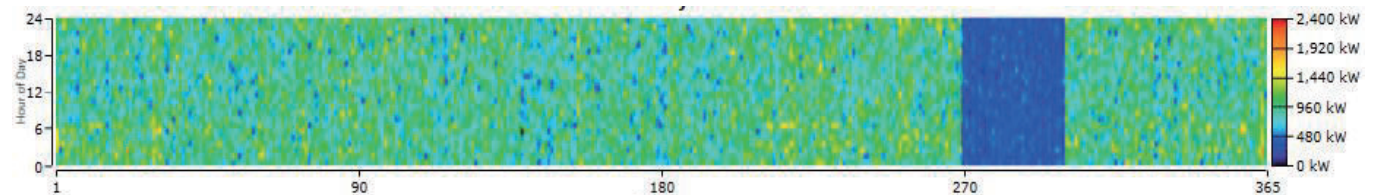


Fig. 3. Yearly load Profile

TABLE I. THE BASE AND SCALE VALUES OF THE PROFILE

Metric	Baseline	Scaled
Average(Kwh/day)	22,763	24,000
Average(KW)	948,49	1000
Peak(KW)	2,003.2	2112.0
Load Factor	0.47	0.47

Table I presents the scale and base values for the load profile. The Homer software uses the scale value for calculations by multiplying each base value by a common factor, ensuring the annual average aligns with the specified value. The load profile decreases and remains constant between 6:00 and 18:00 due to rising temperatures, as the temperate environment aids in maintaining the crude oil's desired temperature. Subsequently, the load profile increases again. During the summer season (February to June), the load profile exhibits a continuous decline, with the highest values recorded at night, as the ambient temperature is not as elevated compared to the winter months.

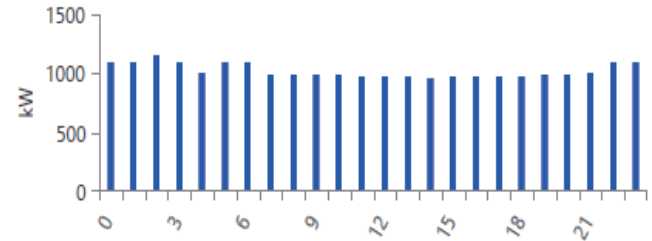


Fig. 1. Daily Load profile

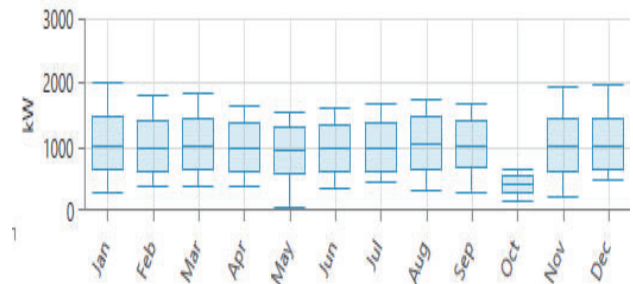


Fig. 2. Seasonal load Profile

B. Wind and Solar Resources Potential:

The annual average temperature is 28.31°C, with March and April recording the highest average temperatures exceeding 30.5°C, while July and August record the lowest monthly averages, below 27°C. During simulations, HOMER incorporates the available ambient temperature data to evaluate the cell temperature, which is subsequently factored into the computation of the photovoltaic (PV) system's output power.

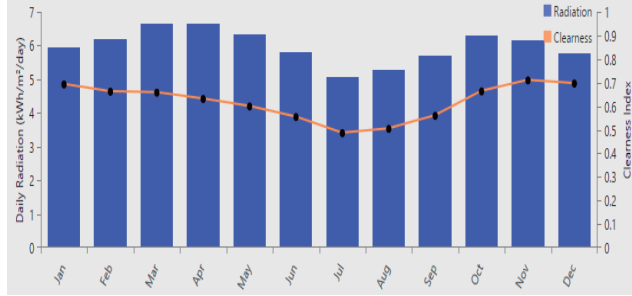


Fig. 4. Monthly average solar global radiation data for the selected location.

Fig. 4 represents the changes in the monthly solar radiation data for the selected location as data obtained from HOMER via NASA satellite database.

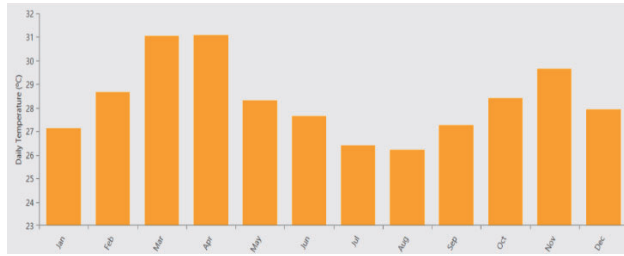


Fig. 5. Monthly average Temperature Data for the selected location

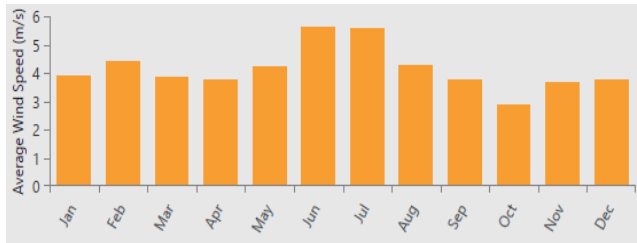


Fig. 6. Average wind speed data in the selected location.

The location has the following potential energies as shown in Fig. 5 and Fig. 6, The data obtained from the National Aeronautics and Space Administration, NASA. The average wind speed is 4.15 m/s, ranging between 3m/s and 5.3 m/s. during the winter months, sunlight is less available due to cloudy sky, but at the same time, wind speed is much higher than at any other time of the year. In summer, wind speed is reduced, but sunshine is high for a longer period of time. As the wind does not blow during the day and the sun does not shine all day, the use of a single source of energy will not be a good choice. Because the optimal operating times of wind systems are different throughout the day and

year than in solar systems, hybrid systems are more suitable to produce the necessary energy.

III. MODELING OF THE SYSTEM COMPONENTS

In this section, various mathematical models are presented to analyze the performance and economics of hybrid power systems (HPS) involving photovoltaic (PV) panels, wind turbines, diesel generators, inverters, and battery storage systems. Each model focuses on specific components and their interactions within the system, including power generation, efficiency, and environmental impact. Additionally, the economic analysis includes key factors such as Net Present Cost (NPC), Cost of Energy (COE), and environmental emissions, providing a comprehensive understanding of system performance and sustainability. The following subsections detail these models and their respective equations.

A. PV and Temperature Model:

PV panels are made up of solar cells in a series or parallel connection and are utilized in a PV system to generate and supply power for different applications. In general, the PV system output power depends on the cell temperature and amount of solar irradiation, as well as the graphical characteristic of an area. In Homer, PV output power is calculated as follows:

$$P_{PV} = Y_{PV} f_{PV} \left(\frac{G_T}{G_{T,STC}} \right) [1 + \alpha_p (T_c - T_{c,STC})]$$

Where Y_{PV} refers to the PV power output under standard test conditions (STC) in KW, f_{PV} represents the PV-de-rating factor (%), G_T is the solar radiation incident on the PV panels in the current time step (KW/m²), $G_{T,STC}$ refers to the incident radiation under standard test conditions (1 KW/m²), α_p is the temperature coefficient of power (%/°C), T_c is the temperature of the PV cell (°C), and $T_{c,STC}$ is the PV cell temperature at STC (25 °C). The derating factor is the decrease in the output if the PV array resulting from temperature losses or any other factors that vary the expected output of the PV system under absolute condition.

B. Wind turbine Modeling:

Wind energy is converted into mechanical power via a wind turbine and then transformed into electrical energy. However, it is necessary to evaluate the past wind power intensity (W/m²) to examine the economic feasibility of a location since the amount of energy captured by the turbine rotor is determined by the wind speed's cubic power. The output power of the wind system is thus calculated as follows:

$$P_{WTG} = \left(\frac{\rho}{\rho_0} \right) \cdot P_{WTG,STP}$$

Where ρ_0 refers to the air density at standard temperature and pressure (STP) (1.225 kg/m³), and $P_{WTG,STP}$ the power output of the wind turbine at STP (KW).

C. Modeling of Diesel generator consumption:

Renewable energy sources are highly dependent on environmental conditions, which greatly influence power generation levels. Hence, they are integrated into a conventional power generation system to supply continuous power output. A diesel generator functions as an uninterrupted source of energy in different hybrid energy systems and is characterized by efficiency and fuel consumption. In this study, the diesel generator system is expected to feed the load based on the dispatch strategy employed. In HOMER, a fuel curve and linear correlation are employed while computing fuel consumption. Fuel consumption is the quantity of fuel utilized to generate electricity, and it is obtained from:

$$F_d = (a.T_d + b.P_d)$$

Where F_d , a , T_d , b , P_d represent the fuel consumption rate of diesel generator (L/h), the coefficient of fuel intercept (L/kWh), the diesel generator capacity, the fuel slope (L/kWh), and the generator output, respectively.

D. Mathematical Model of Inverter:

The inverter is a key power electronic device responsible for converting direct current (DC) from the DC busbar into alternating current (AC) to supply AC loads. The power transiting the inverter to serve the demand is given by:

$$P_{in} = \frac{P_{ch}}{\eta_{in}}$$

Where η_{in} is the inverter efficiency specified by the manufacturer (%), P_{ch} is the hourly demand (w).

E. Mathematical Model of Battery Storage System:

The battery is a storage system required for storing electric power for reliable and efficient utilization of unpredictable RE systems. The storage capacity of the battery is computed based on autonomy days and demand as follows:

$$C_{Bat} = \frac{E_L AD}{\eta_{inv} DOD \eta_{bat}}$$

Where E_L refers to the average load energy per day (KWh/day), AD represents the autonomy days η_{inv} denotes inverter efficiency (95%), DOD represents the depth of discharge (80%), and η_{bat} denotes battery efficiency (85%).

F. Economic Analysis and Modelling:

Economics plays a fundamental role in HOMER by either minimizing or searching for the system with the least net present cost (NPC). HOMER applies the equation below to find the annual real discount rate (or real interest rate).

$$i = \frac{i' - f}{1 + f}$$

where i donates real interest rate, f refers to the expected inflation rate, and i' donates the nominal discount rate.

G. Net Present Cost (NPC):

The NPC consists of the initial cost, replacement cost, fuel cost, cost of grid electricity purchased, as well as the salvage and operation and, maintenance costs. NPC proved to be more reliable as an economic variable due to its mathematical foundation than the levelized electricity cost, which is somewhat arbitrary. The following equation is used to compute NPC:

$$C_{NPC} = \frac{C_{ann,tot}}{CRF(i, n)}$$

Where $C_{ann,tot}$ refers to the total annualized cost (\$/year), n refers to the lifespan of the project (year), i denotes annual real interest rate (%), and CRF , which is a function of i and n , refers to the capital recovery factor:

$$CRF(i, n) = \frac{i(1+i)^n}{(1+i)^n - 1}$$

H. Cost of Energy (COE):

The COE is the average cost per kWh of effective electricity generated by the system, HOMER applies the following expression to evaluate COE:

$$COE = \frac{C_{ann,tot}}{L_{prim} + L_{def} + EI_{grid,sales}}$$

Where L_{prim} and L_{def} represent the overall amounts of load served by the system per year (primary and deferrable load), and $EI_{grid,sales}$ refers to the quantity of electricity sold per year to the utility grid.

I. Environmental analysis and modeling:

The harmful greenhouse gas emissions in the hybrid generation system result from the quantity of fuel intake by the diesel generator and utility grid power consumed. In this study, CO₂ emissions are used to examine the environmental impact of the HPS. The total amount of CO₂ emitted by any hybrid generation system will depend on whether the system includes non-renewable energy sources or not. HOMER computes the emissions factor for individual pollutants before simulation and multiplies this factor by the net annual fuel consumption to determine the annual emissions of each pollutant at the end of the simulation. The equation for computing the amount of hybrid system CO₂ is given as:

$$tCO_2 = 3.667 * m_f * HV_f * CEF_f * X_c$$

where m_f refers to the amount of fuel (liter), HV_f denoted the fuel heating value (MJ/L), CEF_f represents the carbon emission factor (ton carbon/TJ), and X_c is the oxidized carbon fraction.

IV. OPTIMIZED DESIGN OF HYBRID MODEL

In this project, HOMER is used to design the appropriate component where net present cost is the main criteria. Results are compared with the results of Base system which is consist of two diesel generator and Hybrid power system which consist of 1-diesel generator, PV and Wind.

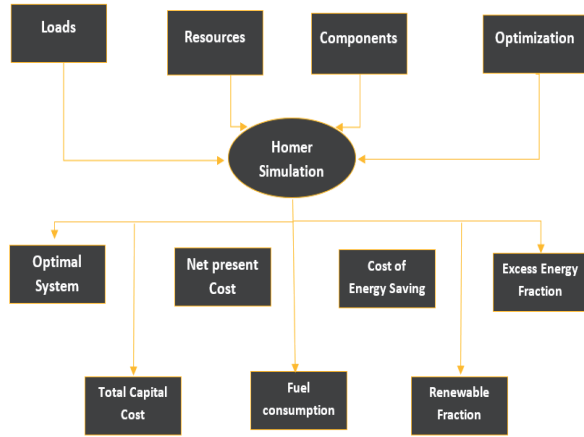


Fig. 7. Architecture Diagram of Homer Simulation

Fig. 7 shows the architecture diagram of HOMER, the software needs the load profile, solar radiation data, wind speed data, and prices of the system components etc.

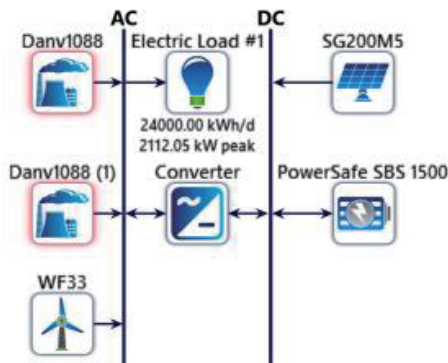


Fig. 8. Schematic of the HES design using HOMER

In the hybrid PV/Wind/Diesel energy system, there are five main components to be considered. They are the generators, PV modules, wind turbine, batteries, and inverter, for the purpose of simulation, the hybrid PV/Diesel energy system was designed in such a way that it consisted of one diesel generator, Photovoltaic array, wind turbine, one inverter and a battery bank (optional), to supply the AC load.

TABLE II. TECHNICAL SPECIFICATION OF THE SYSTEM

Description	Data
PV Technical Specifications	
Panel Type	Flat plate
Total Capacity, (kW)	8647 KW
Manufacture	Pemiar Inc
Lifetime (Years)	30
Derating Factor (%)	80
Efficiency (%)	70
Operating Temperature	25
Wind Turbine Technical Specification	
Supplier/manufacturer	Wind Flow
Rated capacity, (kW)	500 KW
Lifetime, (Years)	20
Hub Height	30 m
Rotor Diameter	33.2 m
Diesel Generators Technical Specifications	
Generator type	AC
Generator capacity, (kW)	1088 KW
Manufacture	Danvest
Lifetime (hours)	64000.00
Fuel cost, (US\$/l)	2.9
Minimum load ratio, (%)	-10
Battery Storage Technical Specifications	
Nominal Capacity	16.7 KWh
Nominal Voltage	12V
Maximum Capacity	1.39E+03 Ah
Manufacture	Energys Powersafe
Lifetime (Years)	15
Throughput (KWh)	17,811.30
Capacity Ratio	0.297
Round Trip Efficiency (%)	97
Maximum Charge Current	1.5E+03 A
Maximum Discharge current	2.3E03A
Maximum Charge Rate (A/Ah)	1
Nominal Capacity	16.7 KWh
Converter Technical Specifications	
Panel Type	Flat plate
capacity, (kW)	1 KW
Manufacture	Sys Converter
Lifetime (Years)	15
Inverter Efficiency (%)	95

V. RESULT AND DISCUSSION

A. Economic Analysis

Following the execution of the model, a range of feasible solutions was obtained, providing key insights into the economic and operational performance of the hybrid energy system. The optimal solution demonstrated a net present cost (NPC) of \$66,083,560, with a capital expenditure (CAPEX) of \$19,181,807.45 per year and an operational cost (OPEX) of \$1,113,712.79 per year. The resulting cost of energy (COE) was calculated at \$0.5838/kWh, which is a significant improvement compared to the base diesel-only system. Detailed cost breakdowns of each system component are provided in Table III, offering a comprehensive overview of the financial distribution between capital investment, operational expenses, and system maintenance.

A comparative analysis was performed between the base system, which comprises two diesel generators (each with a capacity of 1.088 MW), and the proposed hybrid system. The hybrid configuration integrates one diesel generator (1.088 MW), a photovoltaic (PV) array (8647 kW), a wind turbine (500 kW), and a battery storage system with a capacity of 1390 Ah. This optimized hybrid system achieved substantial economic and environmental benefits. Fuel consumption was reduced by 52%, leading to a significant decrease in operational dependency on diesel. Additionally, the total net present cost (NPC) and cost of energy (COE) were reduced by 29%, reflecting improved cost efficiency. Furthermore, operation and maintenance (O&M) expenses saw an 84% reduction, emphasizing the long-term sustainability and lower upkeep costs of the hybrid system.

Table IV summarizes the key performance metrics of the base system versus the hybrid system. The integration of renewable energy sources alongside the conventional diesel generator contributed to a more diversified and resilient energy portfolio. Figure 9 illustrates the lifecycle cost savings generated by the optimized hybrid system over the project's operational lifetime. These savings highlight the economic feasibility and sustainability of incorporating renewable energy sources in regions where diesel generators dominate power generation. The hybrid system not only reduces operational costs but also curbs greenhouse gas emissions, contributing to a cleaner, more sustainable energy future.

TABLE III. COST DATA INPUT OF HOMER SOFTWARE

Component	Capital cost	Replacement cost	Operation cost
Diesel generator	\$ 652,800.00	\$217,600.00	16.320 \$/hr
Wind Turbine	\$8,000,000.00	\$800,000.00	400.00 \$/yr
Photovoltaic cell	\$ 2,000.00	\$ 2,000.00	5.00 \$/yr
Battery	\$ 15,000.00	\$ 15,000.00	20.00 \$/yr
Converter	\$300.00	\$ 300.00	5.00 \$/yr

TABLE IV. COMPARISON BETWEEN THE BASE SYSTEM AND HYBRID SYSTEM

Cost Summary	Base System	Hybrid System
Primary Cost	\$1,305,600.00	\$19,181,807.45
Operating cost (\$/year)	\$7,173,353.00	\$1,113,712.79
Replacement Cost	\$401,703.48	\$4,565,696.34
Levelized COE	\$0.8263	\$0.5838
Fuel Cost	\$89,038,417.94	\$42,170,762.27
Salvage	\$36,306.19	\$948,421.27
Total NPC	\$ 93,573,850.00	\$66,083,560.00

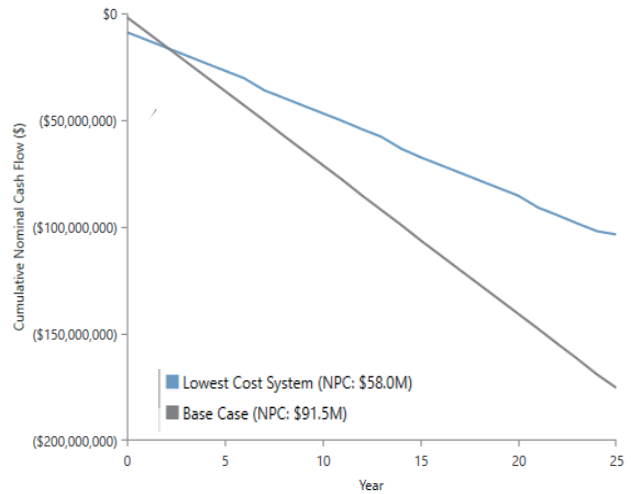


Fig. 9. Cost of savings by the Hybrid System over the project's lifetime

B. Electrical Production

The total electrical production is the total amount of electrical energy produced by the power system in one year. Table V illustrates the sum of the electrical energy produced by all components of the hybrid system. Moreover, Figure 10 demonstrates the monthly electrical production in terms of barograph.

TABLE V. ELECTRICAL PRODUCTION OF EACH ENERGY SOURCE

Production	KWh/year	%
Pv (Peimar)	15,128,687	77.3
Generator (Danvest)	4,333,541	22.1
Wind (wind flow)	118,246	0.604
Total	19,580,474	100

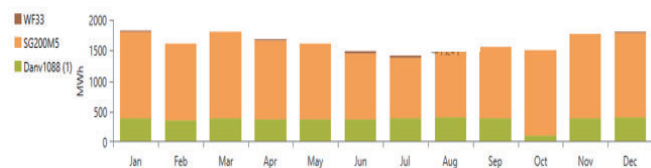


Fig. 10. Monthly Electrical Production

C. Emission Analysis

The utilization of clean sources of energy also results in an additional benefit of reducing in greenhouse gases (GHG) emissions. In the case of the existing system, a total of 6,122,317 KG/year were added into the local atmosphere, while in the case of the hybrid system 2,975,954 KG/year. This illustrates a reduction of 51.4% in total GHG annually of the existing diesel-only system. Over the lifetime of the plant of 25 years a total of 78,659,075 KG/year GHG could be prevented from entering into the local atmosphere, if a carbon incentive of 10 US\$/1000 KG is assumed then a total of 786,600 US\$ Could be earned as carbon benefit which will further reduce the COE. Energy generation from Hydropower systems reduces the emission of CO₂, CO, SO₂, and NO_x into the atmosphere. Table VI shows a summary of emission analysis for the base system and hybrid system. From this emission analysis hybrid system is proved to be the most feasible system and emissions of CO₂, CO, SO₂, and NO_x are less compared to the base system (Diesel Generators).

TABLE VI. AMOUNT OF EMISSIONS

Pollutant	Emission (KG/year)	
	Base System (2- Diesel Generators)	Current System (Pv, wind ,1- Diesel)
Carbon Dioxide	6,068,050	2,949,576
Carbon Monoxide	31,472	15,298
Unburned Hydrocarbons	1,666	810
Particulate matter	271	132
Sulfur dioxide	14,833	7,210
Nitrogen oxides	6,025	2,928

VI. CONCLUSIONS

This study evaluated the economic feasibility of integrating photovoltaic (PV) and wind energy into the existing diesel-based system to meet the energy demands of a selected facility. The HOMER software simulation demonstrated that the hybrid PV/wind/diesel system, combined with battery storage, significantly reduces reliance on diesel as the sole energy source. While the adoption of the PV/wind/diesel hybrid system with battery storage may not drastically lower the total Net Present Cost (NPC) or Cost of Energy (COE), it substantially decreases diesel consumption and associated environmental impacts, such as carbon emissions, contributing to a reduction in the greenhouse effect. Additionally, the study confirmed that the hybrid system becomes even more economically viable with a significant rise in diesel prices. Implementing the proposed hybrid system in the current facility would yield the following key benefits:

- A 29% reduction in the Cost of Energy (COE), decreasing from \$0.8263/kWh to \$0.5838/kWh.
- A 51.4% reduction in total greenhouse gas emissions, from 6,122,317 kg/year to 2,975,954 kg/year.

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